Jenkins Installation on Ubuntu 22.04

Step 1:

Update system

sudo apt update
sudo apt upgrade -y

Step 2: Install Java (Jenkins needs Java)

Jenkins works great with OpenJDK 17 on Ubuntu 22.04.

sudo apt install openjdk-17-jdk -y

java -version

```
root@cvr-ThinkCentre-neo-50t-Gen-3:/home/cvr# java -version
openjdk version "17.0.18" 2026-01-20
OpenJDK Runtime Environment (build 17.0.18+8-Ubuntu-122.04.1)
OpenJDK 64-Bit Server VM (build 17.0.18+8-Ubuntu-122.04.1, mixed mode, sharing)
```

Step 3 : Add Jenkins repository & key

curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee \
  /usr/share/keyrings/jenkins-keyring.asc > /dev/null

echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null

Step 4 : Install Jenkins

sudo apt update
sudo apt install jenkins -y

Step 5 :Start & enable Jenkins

sudo systemctl start jenkins

```
root@cvr-ThinkCentre-neo-50t-Gen-3:/home/cvr# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset:
     Active: active (running) since Mon 2026-02-09 14:48:53 IST; 39s ago
   Main PID: 22223 (java)
      Tasks: 56 (limit: 9163)
     Memory: 1.0G
        CPU: 16.111s
     CGroup: /system.slice/jenkins.service
             └─22223 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/j

Feb 09 14:48:47 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: [LF]> This may also
Feb 09 14:48:47 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: [LF]>
Feb 09 14:48:47 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: [LF]> **************
Feb 09 14:48:47 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: [LF]> **************
Feb 09 14:48:47 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: [LF]> **************
Feb 09 14:48:53 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: 2026-02-09 09:18:53.
Feb 09 14:48:53 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: 2026-02-09 09:18:53.
Feb 09 14:48:53 cvr-ThinkCentre-neo-50t-Gen-3 systemd[1]: Started Jenkins Continuo
Feb 09 14:48:55 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: 2026-02-09 09:18:55.
Feb 09 14:48:55 cvr-ThinkCentre-neo-50t-Gen-3 jenkins[22223]: 2026-02-09 09:18:55.
```

sudo systemctl enable jenkins

Check status:

sudo systemctl status Jenkins

Install Git

```
sudo apt install git -y
```

```
git -version
```

Check where Git is installed
which git

output: /usr/bin/git

Step 6:Allow Jenkins port (if firewall enabled)OPTION

Jenkins uses **port 8080**.

sudo ufw allow 8080
sudo ufw reload

Step 7 Get server ip

ip a

```
root@cvr-ThinkCentre-neo-50t-Gen-3:/home/cvr# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP grou
    link/ether 74:56:3c:b3:ad:ee brd ff:ff:ff:ff:ff:ff
    inet 10.64.21.32/16 brd 10.64.255.255 scope global dynamic noprefixroute enp2s
       valid_lft 85497sec preferred_lft 85497sec
    inet6 fe80::a07:3dab:32b7:6a54/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN
    link/ether 1a:1b:10:b0:70:30 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
       valid_lft forever preferred_lft forever
4: br-acb9d04c34bf: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue sta
    link/ether 5a:22:89:53:81:c6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.49.1/24 brd 192.168.49.255 scope global br-acb9d04c34bf
       valid_lft forever preferred_lft forever
root@cvr-ThinkCentre-neo-50t-Gen-3:/home/cvr# enp2s0
inet 10.64.21.32/16
```

http://10.64.21.32:8080

Step 8 Access Jenkins in browser

Open:

http://<your-server-ip>:8080

http://10.64.21.32:8080

(or http://localhost:8080 if local VM)

Step 9 :Get initial admin password

sudo cat /var/lib/jenkins/secrets/initialAdminPassword

```
root@cvr-ThinkCentre-neo-50t-Gen-3:/home/cvr# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
410172a04f4c43f981f46d52ae23708a
```

Paste it into the Jenkins UI.

Select install suggested plugins

5. Continuous Integration using Jenkins: Configure Jenkins, Jenkins Management.

Step 1: Configure Jenkins (basic setup)

1. Go to **Manage Jenkins** → **Configure System**.
2. Set **Jenkins URL** (it usually auto-fills with your server IP):

http://10.64.21.32:8080/

3. Save changes.

Step 2: Add a Sample Project for CI

We'll create a **simple freestyle project**:

1. Click **New Item**.
2. Enter name: MyFirstCI.
3. Select **Freestyle project** → OK.
4. Scroll to **Source Code Management**:
    - Select **Git**.
    - Repository URL: (you can use a public repo for lab)

      https://github.com/octocat/Hello-World.git
5. Scroll to **Build Triggers**:

- Check **Poll SCM** (for lab) and enter:
  * * * * *
- This polls Git every minute (lab purposes).

6. Scroll to **Build** → Add **Execute shell**:

```
echo "Hello from Jenkins Build"
```

7. Save the project.

Step 3: Run the Build

1. Go to your project → **Build Now**.
2. Click the build number → **Console Output**.
3. You should see:

```
Hello from Jenkins Build
```

Step 4: Manage Jenkins (lab-level)

1. **View Jobs** → See all builds and status.
2. **Configure Jobs** → Change Git repo or commands.
3. **Plugins** → Install useful plugins:
   - Git, GitHub, Pipeline, Slack notifications.
4. **Users** → Add lab users if needed.

Step 5: Pipeline

1. Create a **Pipeline** job.
2. Use simple pipeline script:

```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                echo 'Building the project...'
            }
        }
        stage('Test') {
            steps {
                echo 'Running tests...'
            }
        }
    }
}
```

3. Run → You see stages **Build** → **Test**.

6.Scheduling build Jobs using Jenkins: POLL SCM, Build Periodically.

Scheduling build Jobs using Jenkins

Step 1: new job-freestyle-build steps-echo " This is build job"-apply save
SECONG JOB- new job-freestyle-build steps-echo " This is deploy job"-apply save
3RD JOB- new job Test-freestyle-apply save
4TH JOB- new job release-freestyle-apply save
GOTO BUILD JOB- Configure-postbuild Action-build Other Projects-deploy1 Select Trigger only if build is stable-APPLY SAVE
GOTO DEPLOY JOB- CONFIGURE-POSTBUILD ACTION-BUILD OTHER PROJECTS-TEST1 SELECT Trigger Only If Build Is Stable-apply SAVE
RELEASE JOB- Configure- Build Trigger-select Build After Other Projects Are Built-projects To Watch- Test- Trigger Only If Build Is Stable-apply Save

POLL SCM, Build Periodically.

Jenkins Dashboard

Job Name

Configure

Build Triggers

Now you will see options for scheduling.

case 1: Poll SCM (Build when code changes)

Poll SCM
Schedule

Example 1 – Check every 5 minutes

```
H/5 * * * *
```

Example 2 – Check every 15 minutes

```
H/15 * * * *
```

Example 3 – Every hour

```
H * * * *
```

Save

case 2: Build Periodically (Build at fixed time)

even if there are no changes

Steps

Build periodically

Example 1 – Every 5 minutes

```
H/5 * * * *
```

Example 2 – Every day at 10 AM

```
0 10 * * *
```

Example 3 – Every Sunday at midnight

```
0 0 * * 0
```

Save